

step03_01_data_warehouse

October 8, 2026

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[1]: import duckdb
      from pathlib import Path

      # 1. Establish Directory Paths
      BASE_DIR = Path.cwd().parent
      RESULT_DIR = BASE_DIR / "results"

      # 2. Initialize the Phase 2 OLAP Warehouse Database
      db_path = RESULT_DIR / "step03_01_data_warehouse.duckdb"
      con = duckdb.connect(str(db_path))
      print(f"Connected to OLAP Warehouse: {db_path.name}")

      # 3. Construct the Star Schema View dynamically reading from your CSVs
      star_schema_query = f"""
      CREATE OR REPLACE VIEW olap_star_schema AS

      -- Step A: Construct the Central Fact Table (Merging Delay & Schedule)
      WITH fact_delay AS (
          SELECT
              d.date,
              d.train_no,
              d.station_no,
              d.station_name,
              d.delay,
              s.distance_from_origin,
              s.arrival_day,
              s.arrival_time,
              s.departure_day,
              s.departure_time,
              -- Dynamically resolve operational context based on sequence
              CASE WHEN d.station_no = 1 THEN 'Departure Delay' ELSE 'Arrival Delay' AS
      ↪END AS delay_type
      FROM '{RESULT_DIR.as_posix()}/combined_delay_cleaned.csv' d
      JOIN '{RESULT_DIR.as_posix()}/combined_schedule_imputed.csv' s
          ON d.train_no = s.train_no
          AND d.station_no = s.station_no
      )
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-- Step B: Link to Dimension Tables (The Starburst Branches)
SELECT
    -- Fact Measures & Degenerate Dimension
    f.date,
    f.train_no,
    f.station_name,
    f.station_no,
    f.distance_from_origin,
    f.arrival_day,
    f.arrival_time,
    f.departure_day,
    f.departure_time,
    f.delay AS delay_minutes,
    f.delay_type,

    -- Time Dimension (dim_date)
    dt.year,
    dt.quarter,
    dt.month,
    dt.month_name,
    dt.day_of_month,
    dt.day_name,
    dt.day_of_week,
    dt.is_weekend,

    -- Train Dimension (dim_train_resolved)
    tr.train_name,
    tr.primary_type_code,
    tr.is_premium_special,

    -- Station Dimension (dim_station_resolved)
    st.station_full_name,
    st.station_zone,
    st.station_address

FROM fact_delay f
JOIN '{RESULT_DIR.as_posix()}/dim_date.csv' dt
    ON f.date = dt.date_id
JOIN '{RESULT_DIR.as_posix()}/dim_train_resolved.csv' tr
    ON f.train_no = tr.train_no
JOIN '{RESULT_DIR.as_posix()}/dim_station_resolved.csv' st
    ON f.station_name = st.station_name;
"""

print("Building Star Schema View over Phase 1 results...")
con.execute(star_schema_query)

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print("-> olap_star_schema successfully created.")

# 4. Test the OLAP capabilities (Roll-up Aggregation)
test_cube = """
SELECT
    quarter,
    primary_type_code,
    COUNT(delay_minutes) AS total_delay_events,
    ROUND(AVG(delay_minutes), 2) AS average_delay_minutes
FROM olap_star_schema
GROUP BY quarter, primary_type_code
ORDER BY quarter, total_delay_events DESC
LIMIT 10;
"""

print("\n--- OLAP CUBE TEST (ROLL-UP BY QUARTER & TRAIN TYPE) ---")
results = con.execute(test_cube).df()
print(results.to_string(index=False))

```

Connected to OLAP Warehouse: step03_01_data_warehouse.duckdb
Building Star Schema View over Phase 1 results...
-> olap_star_schema successfully created.

--- OLAP CUBE TEST (ROLL-UP BY QUARTER & TRAIN TYPE) ---

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FloatProgress(value=0.0, layout=Layout(width='auto'),
style=ProgressStyle(bar_color='black'))

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quarter	primary_type_code	total_delay_events	average_delay_minutes
1	PASS-TRAINS	4210568	33.28
1	EXP-TRAINS	3179570	40.23
1	SF-TRAINS	1186109	49.01
1	T18-TRAINS	85694	16.02
1	PRM-TRAINS	36730	104.60
1	RAJ-TRAINS	28282	47.07
1	SHT-TRAINS	24252	16.34
1	GRB-TRAINS	23829	64.50
2	PASS-TRAINS	4518548	30.67
2	EXP-TRAINS	3302774	35.24

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